Linear Algebra and Its Applications

Section 2.2: The Inverse of a Matrix

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

2.2 The inverse of a matrix 3

Practice problem 2.2.P1: Three determinant tests 3

Practice problem 2.2.P2: Detect singularity while seeking an inverse 3

Exercise 2.2.1: An inverse with determinant two 3

Exercise 2.2.2: An inverse with determinant negative one 4

Exercise 2.2.3: Keep track of a negative determinant 4

Exercise 2.2.4: An inverse with fractional entries 4

Exercise 2.2.5: Use the first inverse to solve a system 4

Exercise 2.2.6: Solve using the inverse from Exercise 3 4

Exercise 2.2.7: Four right sides, one set of row operations 5

Exercise 2.2.8: Undo a similarity expression 5

Exercise 2.2.9: True–false tests for inverses 5

Exercise 2.2.10: Five further inverse statements 6
Exercise 2.2.11: Solve a matrix equation and prove uniqueness 6
Exercise 2.2.12: A right inverse of an invertible matrix is its inverse 6
Exercise 2.2.13: Cancellation and the necessity of invertibility 6
Exercise 2.2.14: Cancel on the correct side 6
Exercise 2.2.15: What the augmented reduction computes 6
Exercise 2.2.16: Recover an invertible factor from its product 7
Exercise 2.2.17: The inverse of three factors 7
Exercise 2.2.18: Solve without pretending that factors commute 7
Exercise 2.2.19: An equation with two outside inverses 7
Exercise 2.2.20: An unknown matrix appearing inside an inverse 7
Exercise 2.2.21: Invertibility implies independent columns 8
Exercise 2.2.22: Invertibility implies spanning 8
Exercise 2.2.23: No homogeneous freedom means every column is a pivot 8
Exercise 2.2.24: Every right side means every row has a pivot 8
Exercise 2.2.25: A zero determinant produces a nonzero null vector 8
Exercise 2.2.26: Verify the two-by-two formula 8
Exercise 2.2.27: Elementary matrices select and change rows 9
Exercise 2.2.28: An elementary row replacement and a printed typo 9
Exercise 2.2.29: Find an inverse by the augmented algorithm 9
Exercise 2.2.30: A second augmented inverse calculation 9
Exercise 2.2.31: A full three-by-three inverse 10
Exercise 2.2.32: A failed inverse algorithm is useful information 10
Exercise 2.2.33: Invert a cumulative-sum matrix 10
Exercise 2.2.34: A scaled cumulative-sum matrix 11
Exercise 2.2.35: Find one inverse column without the others 11
Exercise 2.2.36: Two inverse columns in one augmented system 11
Exercise 2.2.37: A rectangular left inverse need not be a right inverse 12
Exercise 2.2.38: A rectangular right inverse 12
Exercise 2.2.39: Convert forces into deflections 12
Exercise 2.2.40: Stiffness and a prescribed third deflection 12
Exercise 2.2.41: Four measured deflections determine four forces 13
Exercise 2.2.42: Prescribe only the second deflection 13

2.2 The inverse of a matrix

An inverse is a *two-sided* undoing operation: for a square matrix, $AB = BA = I$ means $B = A^{-1}$. Multiplication order matters. In particular, $(AB)^{-1} = B^{-1}A^{-1}$ when both inverses exist. For a 2×2 matrix,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad \text{provided } ad - bc \neq 0.$$

For larger matrices, apply the *same row operations* to both halves of $[A \mid I]$. If the left half becomes I, the right half is A^{-1}. If a pivot is missing on the left, the inverse does not exist. To solve $Ax = b$, one usually row reduces $[A \mid b]$ rather than calculating an entire inverse; both routes are explained where requested below.

Practice problem 2.2.P1: Three determinant tests

Compute the scalar $ad - bc$ in each case:

(a) For $\begin{bmatrix} 3 & -9 \\ 2 & 6 \end{bmatrix}$ it is $3 \cdot 6 - (-9)2 = 36 \neq 0$: invertible.

(b) For $\begin{bmatrix} 4 & -9 \\ 0 & 5 \end{bmatrix}$ it is $4 \cdot 5 - (-9)0 = 20 \neq 0$: invertible.

(c) For $\begin{bmatrix} 6 & -9 \\ -4 & 6 \end{bmatrix}$ it is $36 - 36 = 0$: not invertible. Indeed row 2 is $-2/3$ times row 1.

Practice problem 2.2.P2: Detect singularity while seeking an inverse

Begin with the augmented matrix and eliminate its first column:

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ -1 & 5 & 6 & 0 & 1 & 0 \\ 5 & -4 & 5 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - 5R_1]{R_2 \leftarrow R_2 + R_1} \left[\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 6 & 10 & -5 & 0 & 1 \end{array}\right].$$

Now $R_3 \leftarrow R_3 - 2R_2$ makes row 3 equal to $[0\ 0\ 0 \mid -7\ -2\ 1]$. The coefficient half has only two pivots, so it cannot reduce to I_3. Therefore the given matrix has no inverse. A nonzero row in the *right* half does not repair a missing pivot in A.

Exercise 2.2.1: An inverse with determinant two

Here $ad - bc = 8 \cdot 4 - 6 \cdot 5 = 2$. Swap the diagonal entries, negate the other two, and divide by 2:

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 4 & -6 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix}.$$

For example, the first row of AA^{-1} is $(16-15, -24+24) = (1, 0)$, and the second is $(10-10, -15+16) = (0, 1)$. The reverse product is also I_2, either by direct multiplication or the verified 2×2 inverse formula.

Exercise 2.2.2: An inverse with determinant negative one

For $A = \begin{bmatrix} 3 & 2 \\ 8 & 5 \end{bmatrix}$, $ad - bc = 15 - 16 = -1$. Hence

$$A^{-1} = -\begin{bmatrix} 5 & -2 \\ -8 & 3 \end{bmatrix} = \begin{bmatrix} -5 & 2 \\ 8 & -3 \end{bmatrix}.$$

The negative denominator changes *every* sign, not just the diagonal. Multiplication gives diagonal entries $-15 + 16 = 1$ and $16 - 15 = 1$, and off-diagonal entries $6 - 6 = 0$ and $-40 + 40 = 0$.

Exercise 2.2.3: Keep track of a negative determinant

For $A = \begin{bmatrix} 7 & 3 \\ -6 & -3 \end{bmatrix}$, $ad - bc = -21 + 18 = -3$. Therefore

$$A^{-1} = -\frac{1}{3}\begin{bmatrix} -3 & -3 \\ 6 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -2 & -7/3 \end{bmatrix}.$$

As a check, AA^{-1} has rows $(7 - 6, 7 - 7) = (1, 0)$ and $(-6 + 6, -6 + 7) = (0, 1)$.

Exercise 2.2.4: An inverse with fractional entries

The determinant is $2(-6) - (-4)4 = 4 \neq 0$, so

$$A^{-1} = \frac{1}{4}\begin{bmatrix} -6 & 4 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} -3/2 & 1 \\ -1 & 1/2 \end{bmatrix}.$$

Fractions cause no problem: inverse entries need not be integers even when all entries of the original matrix are integers.

Exercise 2.2.5: Use the first inverse to solve a system

Multiply the right side by the inverse from Exercise 1:

$$x = A^{-1}b = \begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 + 3 \\ -5 - 4 \end{bmatrix} = \begin{bmatrix} 7 \\ -9 \end{bmatrix}.$$

Check in the original equations: $8(7) + 6(-9) = 2$ and $5(7) + 4(-9) = -1$. Since A is invertible, this solution is unique.

Exercise 2.2.6: Solve using the inverse from Exercise 3

$$x = \begin{bmatrix} 1 & 1 \\ -2 & -7/3 \end{bmatrix}\begin{bmatrix} -9 \\ 4 \end{bmatrix} = \begin{bmatrix} -5 \\ 18 - 28/3 \end{bmatrix} = \begin{bmatrix} -5 \\ 26/3 \end{bmatrix}.$$

Substitution gives $7(-5) + 3(26/3) = -9$ and $-6(-5) - 3(26/3) = 4$. Both equations, including their signs, are satisfied.

Exercise 2.2.7: Four right sides, one set of row operations

(a) Compute the inverse and multiply each right side.

The determinant of $A = \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix}$ is $12 - 10 = 2$, so $A^{-1} = \begin{bmatrix} 6 & -1 \\ -5/2 & 1/2 \end{bmatrix}$. For any $b = (c, d)^T$, its solution is $(6c - d, (-5c + d)/2)^T$. Applying this formula to the four supplied columns gives, in order,

$$(-9, 4)^T, \quad (11, -5)^T, \quad (6, -2)^T, \quad (13, -5)^T.$$

(b) Solve the four systems together.

Use $R_2 \leftarrow R_2 - 5R_1$, then $R_2 \leftarrow R_2/2$, then $R_1 \leftarrow R_1 - 2R_2$:

$$\left[\begin{array}{cc|cccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 5 & 12 & 3 & -5 & 6 & 5 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 0 & 2 & 8 & -10 & -4 & -10 \end{array}\right]$$

$$\longrightarrow \left[\begin{array}{cc|cccc} 1 & 0 & -9 & 11 & 6 & 13 \\ 0 & 1 & 4 & -5 & -2 & -5 \end{array}\right].$$

Every column on the right has undergone the same operations as its own separate augmented system, so the final block lists exactly the four solutions.

Exercise 2.2.8: Undo a similarity expression

From $A = PBP^{-1}$, multiply on the left by P^{-1} and on the right by P:

$$P^{-1}AP = P^{-1}(PBP^{-1})P = (P^{-1}P)B(P^{-1}P) = B.$$

Thus $\boxed{B = P^{-1}AP}$. Putting both new factors on the same side would not cancel the original factors.

Exercise 2.2.9: True–false tests for inverses

(a) **True.** The definition of inverse requires $AB = BA = I$. For square matrices one equality eventually implies the other, but both equalities are still true when B is the inverse.

(b) **False.** The correct order is $(AB)^{-1} = B^{-1}A^{-1}$. For instance take $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \operatorname{diag}(1, 2)$. Then $A^{-1}B^{-1}$ is $\begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix}$, whereas the actual inverse is $\begin{bmatrix} 1 & -1 \\ 0 & 1/2 \end{bmatrix}$.

(c) **False.** The determinant is $ad - bc$, not $ab - cd$. The singular matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has $ab - cd = 2 - 8 = -6 \neq 0$, disproving the assertion.

(d) **True.** For each b, $x = A^{-1}b$ satisfies $Ax = AA^{-1}b = b$.

(e) **True.** Reverse the underlying row operation: a swap undoes itself, nonzero scaling is undone by its reciprocal, and adding r times another row is undone by adding $-r$ times that row. The corresponding elementary matrix is the inverse.

Exercise 2.2.10: Five further inverse statements

(a) **False.** Operations taking A to I collectively multiply by A^{-1}. Applied to A^{-1} they give A^{-2}, usually not I. For $A = 2I$, dividing rows by 2 takes $A^{-1} = I/2$ to $I/4$.

(b) **True.** $A^{-1}A = AA^{-1} = I$ says precisely that A is the inverse of A^{-1}.

(c) **False.** The product is invertible, but inverses occur in reverse order; the counterexample in Exercise 9(b) applies.

(d) **True.** Solvability for each e_j lets us place solutions in columns of D with $AD = I$. Alternatively, every $b = \sum_j b_j e_j$ is an output of A by linearity, so A has a pivot in every row. Since A is square, its reduction is I, proving invertibility.

(e) **True.** This is the row-reduction characterization: applying the same operations to I produces the inverse.

Exercise 2.2.11: Solve a matrix equation and prove uniqueness

If $AX = B$, multiply on the left by A^{-1} to obtain $X = A^{-1}B$. This proves every solution must have that form. Conversely, $A(A^{-1}B) = (AA^{-1})B = B$, so that form really is a solution. The two arguments together prove existence and uniqueness. Shapes agree: $A^{-1}B$ is $n \times p$, just as X must be.

Exercise 2.2.12: A right inverse of an invertible matrix is its inverse

Multiply $AD = I$ on the left by A^{-1}:

$$D = (A^{-1}A)D = A^{-1}(AD) = A^{-1}I = A^{-1}.$$

The existing inverse of A justifies the cancellation; it is not a general rule for arbitrary rectangular or singular matrices.

Exercise 2.2.13: Cancellation and the necessity of invertibility

If $AB = AC$ and A is invertible, multiply both sides on the left by A^{-1} to get $B = C$. Without invertibility this need not hold. For example, $A = \operatorname{diag}(1, 0)$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, and $C = 0$ satisfy $AB = 0 = AC$ but $B \neq C$. Thus the hypothesis performs real work.

Exercise 2.2.14: Cancel on the correct side

Starting from $(B - C)D = 0$, multiply on the *right* by D^{-1}: $B - C = (B - C)(DD^{-1}) = 0D^{-1} = 0$. Therefore $B = C$. Left multiplication would not put D next to its inverse.

Exercise 2.2.15: What the augmented reduction computes

Let the row operations be represented by elementary matrices $E_1, \ldots, E_k$, applied in that order. Their combined multiplier is $E = E_k \cdots E_1$. Reducing $[A \mid B]$ to $[I \mid X]$ means $EA = I$ and $EB = X$.

Since A is invertible, $E = EAA^{-1} = A^{-1}$, whence $X = A^{-1}B$. The right block has been transformed along with A, avoiding a separate full inverse calculation.

Exercise 2.2.16: Recover an invertible factor from its product

Put $C = AB$, which is invertible by hypothesis. Right multiplication by B^{-1} gives $A = CB^{-1}$. This is a product of two matrices whose inverses are known: C^{-1} and B. Indeed BC^{-1} is an inverse of A, because

$$(CB^{-1})(BC^{-1}) = I, \qquad (BC^{-1})(CB^{-1}) = I.$$

Thus A is invertible without assuming the desired result in advance.

Exercise 2.2.17: The inverse of three factors

Choose $D = C^{-1}B^{-1}A^{-1}$. Associativity allows the adjacent cancellations

$$(ABC)D = AB(CC^{-1})B^{-1}A^{-1} = AA^{-1} = I,$$

$$D(ABC) = C^{-1}B^{-1}(A^{-1}A)BC = C^{-1}C = I.$$

Both sides have been checked, so D is an inverse of ABC.

Exercise 2.2.18: Solve without pretending that factors commute

From $AB = BC$, right-multiply by B^{-1}: $\boxed{A = BCB^{-1}}$. Substitution checks $(BCB^{-1})B = BC$. In general BCB^{-1} is not C; cancelling the two B factors across C would be invalid.

Exercise 2.2.19: An equation with two outside inverses

Multiply $C^{-1}(A+X)B^{-1} = I$ first on the left by C, then on the right by B. This gives $A+X = CB$, hence $\boxed{X = CB - A}$. Substitution gives $C^{-1}(CB)B^{-1} = I$, proving that a solution always exists under the stated hypotheses. The rearrangement also proves uniqueness.

Exercise 2.2.20: An unknown matrix appearing inside an inverse

(a) First prove that the required inverse exists.

From $(A - AX)^{-1} = X^{-1}B$, left multiplication by X gives $B = X(A - AX)^{-1}$. Both factors on the right are invertible by hypothesis, so B is invertible.

(b) Remove the inverse, then collect the unknown on one side.

The same equality gives $B(A - AX) = X$, or $BA - BAX = X$. Therefore

$$(I + BA)X = BA.$$

This also shows $I + BA = BAX^{-1}$, a product of invertible matrices, so the next inverse is justified. Left multiplication yields

$$\boxed{X = (I + BA)^{-1}BA}.$$

The original assumptions already require an admissible solution X; we used them to prove that every inverse in this expression exists.

Exercise 2.2.21: Invertibility implies independent columns

Suppose a combination of the columns is zero, equivalently $Ax = 0$. Multiplying by A^{-1} gives $x = 0$. Thus all coefficients in the combination are zero, which is exactly linear independence.

Exercise 2.2.22: Invertibility implies spanning

For every $b \in \mathbb{R}^n$, the vector $x = A^{-1}b$ satisfies $Ax = b$. But Ax is the linear combination of A's columns with coefficients from x. Thus every b is such a combination: the columns span $\mathbb{R}^n$.

Exercise 2.2.23: No homogeneous freedom means every column is a pivot

A nonpivot column corresponds to a free variable and would produce a nontrivial solution of $Ax = 0$. The hypothesis rules out that possibility, so all n columns are pivot columns. In an $n \times n$ matrix this means all rows have pivots too. Its reduced echelon form has a leading 1 in each column and zeros elsewhere in those columns, hence is I_n. The inverse algorithm then proves that A is invertible.

Exercise 2.2.24: Every right side means every row has a pivot

If a coefficient row lacked a pivot, some right side would produce a contradictory augmented row $[0 \ \cdots \ 0 \mid 1]$. Consistency for every right side therefore forces n pivot rows. With only n columns there are also n pivot columns, so the reduced echelon form is I_n. Again the inverse algorithm proves invertibility.

Exercise 2.2.25: A zero determinant produces a nonzero null vector

What we must show. Prove singularity directly when $ad - bc = 0$. If $(a, b) \neq (0, 0)$, use $x = (-b, a)^T \neq 0$. Multiplication gives $Ax = (-ab + ba, -bc + da)^T = 0$. If $a = b = 0$ but $(c, d) \neq (0, 0)$, use $x = (-d, c)^T \neq 0$; both rows again give zero. If all four entries vanish, any nonzero x, such as $(1, 0)^T$, works. Thus both $x = 0$ and a nonzero x solve $Ax = 0$. An inverse would imply $x = A^{-1}0 = 0$, contradicting the nonzero solution. Hence A is not invertible.

Exercise 2.2.26: Verify the two-by-two formula

Let $\Delta = ad - bc \neq 0$ and $B = \Delta^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Then

$$AB = \frac{1}{\Delta} \begin{bmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{bmatrix} = I_2,$$

$$BA = \frac{1}{\Delta} \begin{bmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{bmatrix} = I_2.$$

Division by Δ is legal by hypothesis. Both products equal the identity, so $B = A^{-1}$ by definition.

Exercise 2.2.27: Elementary matrices select and change rows

(a) Row i of I_3 has a 1 in position i and zeros elsewhere. Consequently $(\text{row}_i\, I_3)A$ is 1 times row i of A plus zero times the other rows: it equals $\text{row}_i\, A$.

(b) Set $E = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The three rows of EA are, by part (a), rows $2, 1, 3$ of A. Thus it swaps rows 1 and 2, exactly as the same operation on I_3 does.

(c) Set $E = \text{diag}(1, 1, 5)$. Its rows select row 1 of A, row 2 of A, and five times row 3 of A. Thus EA performs the desired scaling, and E is obtained by scaling row 3 of I_3 by 5.

Exercise 2.2.28: An elementary row replacement and a printed typo

The intended elementary matrix is

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Its rows show that EA has rows $\text{row}_1\, A$, $-3\,\text{row}_1\, A + \text{row}_2\, A$, and $\text{row}_3\, A$, as required. To construct E from I, the replacement must be $\text{row}_2(I) \leftarrow \text{row}_2(I) - 3\,\text{row}_1(I)$. The supplied edition prints $\text{row}_1(A)$ in that last construction. Read literally that is incorrect: it would make E depend on A and need not be the elementary matrix for this operation. The computation above states and verifies the corrected version explicitly.

Exercise 2.2.29: Find an inverse by the augmented algorithm

Start with $[A \mid I]$ and perform the indicated operations:

$$\left[\begin{array}{cc|cc} 1 & -3 & 1 & 0 \\ 4 & -9 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 4R_1} \left[\begin{array}{cc|cc} 1 & -3 & 1 & 0 \\ 0 & 3 & -4 & 1 \end{array}\right].$$

Divide row 2 by 3, then add three times the new row 2 to row 1. The result is $\left[\begin{array}{cc|cc} 1 & 0 & -3 & 1 \\ 0 & 1 & -4/3 & 1/3 \end{array}\right]$.

Thus $A^{-1} = \begin{bmatrix} -3 & 1 \\ -4/3 & 1/3 \end{bmatrix}$.

Exercise 2.2.30: A second augmented inverse calculation

For $[A \mid I]$, first use $R_1 \leftarrow R_1/3$, then $R_2 \leftarrow R_2 - 4R_1$:

$$\left[\begin{array}{cc|cc} 3 & 6 & 1 & 0 \\ 4 & 7 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cc} 1 & 2 & 1/3 & 0 \\ 0 & -1 & -4/3 & 1 \end{array}\right].$$

Negate row 2 and replace row 1 by $R_1 - 2R_2$. This gives

$$[I \mid A^{-1}] = \left[\begin{array}{cc|cc} 1 & 0 & -7/3 & 2 \\ 0 & 1 & 4/3 & -1 \end{array}\right].$$

Both left-hand pivots exist, so the inverse is the displayed right block.

Exercise 2.2.31: A full three-by-three inverse

Eliminate below the first pivot of $[A \mid I]$ with $R_2 \leftarrow R_2 + 3R_1$, $R_3 \leftarrow R_3 - 2R_1$, then $R_3 \leftarrow R_3 + 3R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array}\right].$$

Next divide row 3 by 2 and add twice this new row to each of rows 1 and 2. The coefficient block is now I_3, and the right block is

$$\boxed{A^{-1} = \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ 7/2 & 3/2 & 1/2 \end{bmatrix}}.$$

For instance the first row $[1, 0, -2]$ of A times this block is $[8, 3, 1] - 2[7/2, 3/2, 1/2] = [1, 0, 0]$; the other two rows similarly check the identity. Exact multiplication verifies all nine entries.

Exercise 2.2.32: A failed inverse algorithm is useful information

Apply $R_2 \leftarrow R_2 + 4R_1$ and $R_3 \leftarrow R_3 + 2R_1$ to $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ -4 & -7 & 3 & 0 & 1 & 0 \\ -2 & -6 & 4 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 4 & 1 & 0 \\ 0 & -2 & 2 & 2 & 0 & 1 \end{array}\right].$$

Now $R_3 \leftarrow R_3 + 2R_2$ gives a zero coefficient row, with right block row $[10, 2, 1]$. The coefficient block has only two pivots and cannot become I_3. Therefore the inverse does not exist. Stopping here is a conclusion about singularity, not an unfinished calculation.

Exercise 2.2.33: Invert a cumulative-sum matrix

For the 3×3 lower-triangular matrix of ones, apply $R_3 \leftarrow R_3 - R_2$ first and then $R_2 \leftarrow R_2 - R_1$ to $[A \mid I]$. The order ensures that the subtracted rows are still the original rows. The left block becomes I and the right becomes

$$B_3 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

For the 4×4 case, work upward with $R_4 \leftarrow R_4 - R_3$, $R_3 \leftarrow R_3 - R_2$, $R_2 \leftarrow R_2 - R_1$:

$$B_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

In general B has 1 on the diagonal, -1 immediately below it, and zero elsewhere. For $j < n$, column j of AB is $Ae_j - Ae_{j+1}$. The two cumulative columns agree below row j, and differ by 1 exactly in row j, so this is e_j. The last column is $Ae_n = e_n$. Thus $AB = I_n$; the same upward row-reduction construction shows B is indeed the inverse for every n.

Exercise 2.2.34: A scaled cumulative-sum matrix

Let L be the lower-triangular ones matrix and $D = \operatorname{diag}(1, 2, \ldots, n)$. The given A is DL, because row i consists of i's up through its diagonal entry. Thus its inverse is $B = L^{-1}D^{-1}$. Explicitly,

$$b_{ii} = 1/i, \qquad b_{i,i-1} = -1/(i-1) \ (i \geq 2),$$

with every other entry zero. For example, $B_3 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/2 & 0 \\ 0 & -1/2 & 1/3 \end{bmatrix}$; B_4 appends a last row $(0, 0, -1/3, 1/4)$ to the corresponding four-column pattern. For $j < n$, column j of AB is $(Ae_j - Ae_{j+1})/j = e_j$, since the two columns differ only in row j, where their difference is j. The last column is $Ae_n/n = e_n$. Hence $AB = I_n$, verifying the formula.

Exercise 2.2.35: Find one inverse column without the others

The third column z of A^{-1} satisfies $Az = e_3$, since $AA^{-1} = I$. Its equations are

$$-z_1 - 7z_2 - 3z_3 = 0, \quad 2z_1 + 15z_2 + 6z_3 = 0, \quad z_1 + 3z_2 + 2z_3 = 1.$$

Add twice the first equation to the second: $z_2 = 0$. Add the first to the third: $-4z_2 - z_3 = 1$, so $z_3 = -1$. The first then gives $z_1 = 3$. Thus the requested column is $\boxed{(3, 0, -1)^T}$. Substitution verifies all three equations. The coefficient matrix has pivots in all three columns by this elimination, so the use of its inverse is justified.

Exercise 2.2.36: Two inverse columns in one augmented system

Solve $Az = e_2$ and $Aw = e_3$ together. No first inverse column is needed. Start with $[A \mid e_2 \ e_3]$. To avoid early fractions, multiply row 2 by 25 and add 536 times row 1; multiply row 3 by 25 and add 154 times row 1. These are compositions of legal elementary operations:

$$\left[\begin{array}{ccc|cc} -25 & -9 & -27 & 0 & 0 \\ 536 & 185 & 537 & 1 & 0 \\ 154 & 52 & 143 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|cc} -25 & -9 & -27 & 0 & 0 \\ 0 & -199 & -1047 & 25 & 0 \\ 0 & -86 & -583 & 0 & 25 \end{array}\right].$$

Replacing row 3 by $199R_3 - 86R_2$ gives its coefficient row $[0, 0, -25975]$ and right side $[-2150, 4975]$. Back-substitution gives

$$z = \frac{1}{1039}\begin{bmatrix} 117 \\ -583 \\ 86 \end{bmatrix}, \qquad w = \frac{1}{1039}\begin{bmatrix} -162 \\ 1047 \\ -199 \end{bmatrix}.$$

For example $z_3 = 2150/25975 = 86/1039$; the preceding row determines z_2, and the first row determines z_1. The same steps give w. All three triangular pivots are nonzero; exact substitution gives $Az = e_2$ and $Aw = e_3$ and verifies both requested columns.

Exercise 2.2.37: A rectangular left inverse need not be a right inverse

For the supplied 3×2 matrix, choose

$$C = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \end{bmatrix}.$$

All entries are allowed. In CA, its first row gives $[1,2] + [1,3] - [1,5] = [1,0]$; the second gives $-[1,2] + [1,3] = [0,1]$. Thus $CA = I_2$. In the other order,

$$AC = \begin{bmatrix} -1 & 3 & -1 \\ -2 & 4 & -1 \\ -4 & 6 & -1 \end{bmatrix} \neq I_3.$$

The unequal dimensions prevent a general reversal of the identity.

Exercise 2.2.38: A rectangular right inverse

The first and fourth columns of the given A are e_1 and e_2. Select them by taking

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Its entries are only $0, 1$, and $AD = [e_1\ e_2] = I_2$. There cannot be a 4×2 matrix C with $CA = I_4$. Indeed the nonzero vector $(0,1,1,0)^T$ satisfies $Ax = 0$. If $CA = I_4$, then $x = CAx = C0 = 0$, a contradiction.

Exercise 2.2.39: Convert forces into deflections

The model is $y = Df$: flexibility times force gives deflection. With $f = (40, 50, 30)^T$ pounds, the three entries are

$$y_1 = .011(40) + .003(50) + .001(30) = .62,$$

$$y_2 = .003(40) + .009(50) + .003(30) = .66, \qquad y_3 = .001(40) + .003(50) + .011(30) = .52.$$

Thus the deflections are $\boxed{(.62, .66, .52)^T}$ inches. Multiplying units, inches per pound times pounds gives inches.

Exercise 2.2.40: Stiffness and a prescribed third deflection

Row reduction, or exact rational arithmetic on the decimal entries, gives

$$D^{-1} = \begin{bmatrix} 100 & -100/3 & 0 \\ -100/3 & 400/3 & -100/3 \\ 0 & -100/3 & 100 \end{bmatrix}.$$

To check rather than trust the calculation, write $D = 10^{-3}\begin{bmatrix}11 & 3 & 1\\ 3 & 9 & 3\\ 1 & 3 & 11\end{bmatrix}$. Its first row times the displayed stiffness matrix gives $[1100-100, -1100/3+400-100/3, -100+100] = [1000, 0, 0]$; the other rows give $[0, 1000, 0]$ and $[0, 0, 1000]$. The factor 10^{-3} therefore makes the product I. For $y = .04e_3$, use only column 3 of D^{-1}:

$$f = D^{-1}y = .04(0, -100/3, 100)^T = (0, -4/3, 4)^T \text{ pounds.}$$

The negative second force points opposite to the chosen positive direction; it is needed to keep the second deflection at zero.

Exercise 2.2.41: Four measured deflections determine four forces

Let $S = 1000D$, an integer matrix, and use the measured y in centimeters. Then $Df = y$ is equivalent to

$$\begin{bmatrix}13 & 5 & 2 & 1\\ 5 & 10 & 4 & 2\\ 2 & 4 & 10 & 5\\ 1 & 2 & 5 & 13\end{bmatrix} f = \begin{bmatrix}70\\ 120\\ 160\\ 120\end{bmatrix}.$$

Exact row reduction produces

$$[I \mid f], \qquad \boxed{f = (20/21, 130/21, 80/7, 80/21)^T} \text{ newtons.}$$

For a transparent check with integers, write $f = (20, 130, 240, 80)^T/21$. Multiplying by S gives $(1470, 2520, 3360, 2520)^T/21 = (70, 120, 160, 120)^T$, which verifies all four equations. The inverse used in this computation is

$$D^{-1} = \frac{1000}{21}\begin{bmatrix}2 & -1 & 0 & 0\\ -1 & 3 & -1 & 0\\ 0 & -1 & 3 & -1\\ 0 & 0 & -1 & 2\end{bmatrix}.$$

Multiplication by the given D gives I_4, so the forces are unique. The form of this inverse makes the next requested calculation especially simple.

Exercise 2.2.42: Prescribe only the second deflection

A unit deflection e_2 requires $D^{-1}e_2$, precisely column 2 of the stiffness matrix. By linearity, a deflection $.22e_2$ requires .22 times that column. Using the inverse from Exercise 41,

$$f = .22\frac{1000}{21}(-1, 3, -1, 0)^T = \boxed{(-220/21, 220/7, -220/21, 0)^T} \text{ newtons.}$$

Multiplication by D gives $(0, .22, 0, 0)^T$ centimeters. The last force is zero, while two forces are negative: forcing a single deflection does not mean that only one applied force is needed.